Advanced Reactor Fuel Cycles Molten Salt Reactor Design

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1 Brief History

The basis for this quick synopsis comes from [1]. Research into Molten Salt Reactors (MSRs) began in earnest with the Aircraft Reactor Experiment (ARE), with experiments shared between Oak Ridge National Laboratory (ORNL) and what is now Idaho National Laboratory (INL). The ARE used NaF-ZrF_4-UF_4 (53-41-6 mol%) as fuel. The reactor was moderated by beryllium oxide (BeO), used liquid sodium as the secondary coolant, and had a peak temperature around 860 °C. The ARE managed to generate 100 MWh over the course of nine days in 1954. There was another reactor that went critical at ORNL in 1957 called the Pratt and Whitney Aircraft Reactor-1 (PWAR-1). Although it produced essentially no power, the PWAR-1 is still one of only three critical MSRs ever built.

The most successful MSRwas the Molten Salt Reactor Experiment (MSRE) which went critical in 1965 and operated for four years at ORNL. The MSRE fuel was LiF-BeF_2-ZrF_4-UF_4 (65-29-5-1 mol%) with graphite core moderation. The secondary coolant was 2LiF-BeF_2 (also known as FLiBe). Operating temperatures went as high as 650 °C. MSRE neutronics were meant to mimic that of an epithermal thorium molten salt breeder reactor (MSBR); however, for the MSRE the expensive breeding blanket was sacrificed in favor of neutron measurements. At full design power, the MSRE operated at 7.4 MW_{th}. Over the course of the late 1960s the MSRE operated at full capacity for an equivalent of 1.5 years.

The logical follow-on from the MSRE was the MSBR; however, the MSBR was never constructed. Funding for the MSRprogram was cut in favor of the Liquid-Metal-cooled Fast Breeder Reactor (LMFBR) program. Consequently, as of today, the ARE, PWAR-1, and MSRE are the only critical MSRs ever operated.

2 MSRE

2.1 Salt properties

The melting point of FLiBe is 459 °C and the boiling point is 1430 °C [3].

	Fuel Salt			
	Thorium–Uranium	Highly Enriched Uranium	Partially Enriched Uranium	Flush and Coolant Salt
Composition, mole %				
LiF (99.99+% Li^7)	70	66.8	65	66
BeF_2	23.6	29	29.1	34
ZrF_4	5	4	5	--
ThF_4	1	0	0	--
UF_4	0.4	0.2	0.9	--
Physical Properties, at	1200°F	1200°F	1200°F	1060°F
Density, lb/ft^3, at 1200°F	140	130	134	120
Viscosity, lb/ft-hr	18	17	20	24
Heat Capacity, Btu/lb-°F	0.45 (at 1200°F)	0.48	0.47	0.53 (at 1200°F)
Thermal Conductivity, Btu/hr^2 (°F/ft)	3.21	3.2	3.2	3.5
Liquidus Temperature, °F	840	840	840	850

Figure 1: Composition and properties of fuel, flush, and coolant salts [2].

2.2 Layout & Equipment

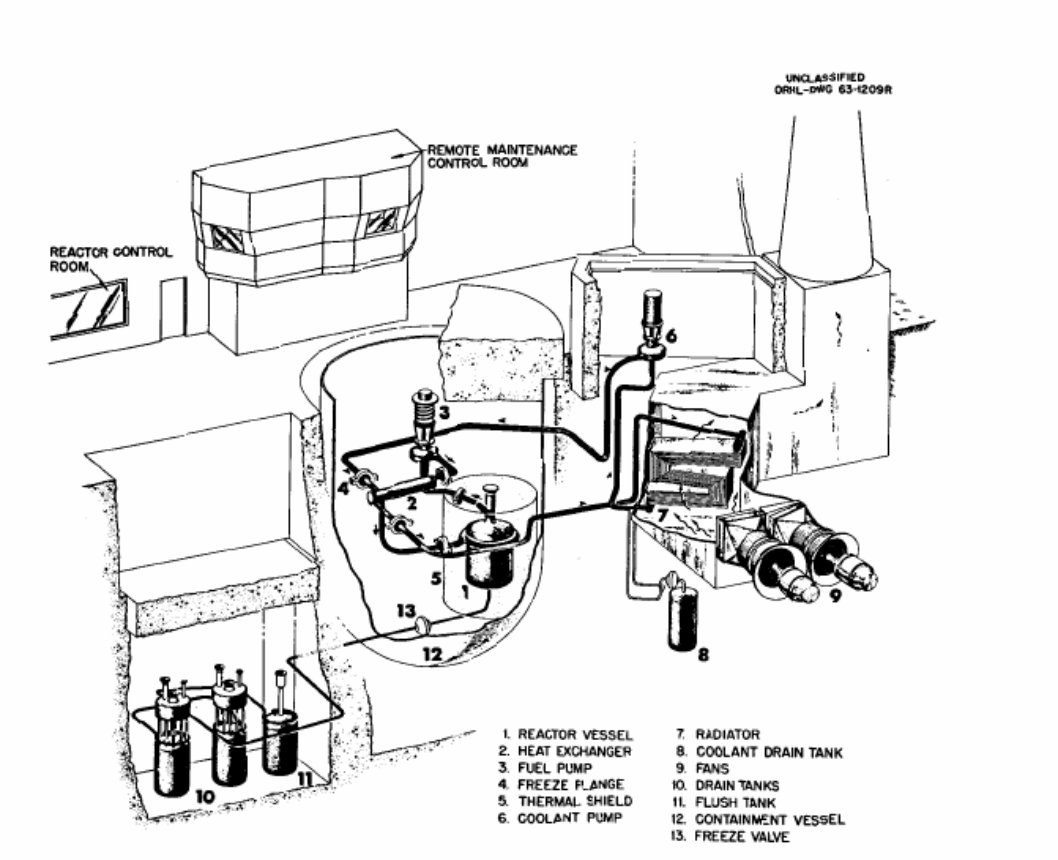

Figure 2: Macroscopic layout of the MSRE [2]

2.2.1 Reactor vessel

A cutaway diagram of the MSRE reactor is shown in fig. 3. Fuel enters through the inlet and is distributed evenly along the outside of the reactor vessel in a small annular region of 1 inch width [2]. The fluid fuel then flows turbulently downward, cooling the reactor vessel to within 5°F of the entering bulk salt temperature (1175 °F) [2]. This down-flowing region is typically referred to as the *downcomer*. The fuel salt is collected in the lower plenum (lower *space*) of the reactor where straightening vanes remove the salt's rotational motion. The salt then flows upward through approximately 1140 0.4-inch by 1.2-inch fuel channels that are formed by machining grooves in the sides of the graphite moderator columns [2]; these are the regions where fission occurs. The graphite columns themselves have 2-inch by 2-inch square cross sections and are 67 inches in length. Nominal core volume is 90 ft^3, with 70 ft^3 occupied by graphite and 20 ft^3 occupied by fuel [2]. After flowing through the core channels, the fuel collects in the upper plenum before exiting through the outlet. At 10 MW, 0.6 MW of heat is generated in the graphite, 1.4 MW heat is generated in the fuel outside the nominal core, and 8.0 MW is generated in the fuel within the core [2]. I'm very curious how those numbers were determined.

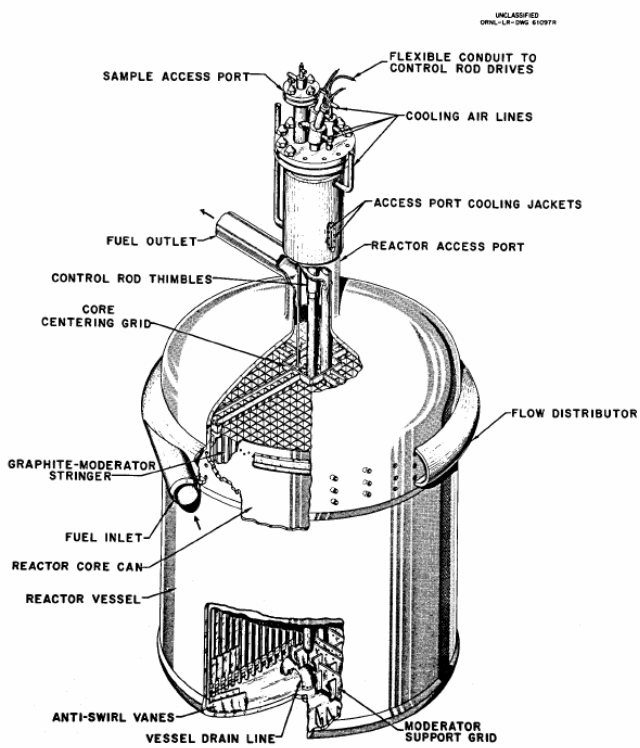

Fig. 2.2. Reactor Vessel.

Figure 3: MSRE reactor vessel [2].

Flow in fuel channels is laminar; because both the graphite and fueld have good thermal conductivities, the maximum temperature of the graphite is only about 60 °F above the mixed mean temperature of the adjacent fuel. Average and maximum temperatures of graphite are 1255 °F and 1300 °F respectively. The temperature of the fuel leaving the hottest channel in the core is about 1260 °F. The fuel leaves the top of the reactor at 7 psig and an average temperature of 1225 °F [2].

Though studies were conducted with the MSRE core surrounded by varying thicknesses of reflector [4], the experimental design was bare. This type of reactor design lends itself most naturally to a vacuum boundary

condition for neutrons (zero incoming flux) like that used in [5].

2.2.2 Heat Exchanger

As mentioned in section 2.2.1, the fuel salt enters the reactor at 1175 °F and leaves at 1225 °F. In order to cool the fuel salt back down to its reactor inlet temperature, the coolant salt flows through the heat exchanger at 850 gallons per minute (gpm) with inlet and outlet temperatures of 1025 °F and 1100 °F respectively [2]. A summary of some of the important operating temperatures in the MSRE is given in table 1.

Material & Location	Temp (°F)	Temp (K)
Fuel salt reactor inlet/heatx outlet	1175	908
Fuel salt reactor average outlet/heatx inlet	1225	936
Fuel salt reactor max outlet	1260	955
Coolant salt heatx inlet	1025	825
Coolant salt heatx outlet	1100	866
FLiBe boiling point	2606	1703
Graphite core average	1255	953
Graphite core maximum	1300	978

Table 1: Some important operating temperatures in the MSRE

2.3 Reactivity considerations

The MSRE has a negative temperature coefficient of $6.4 - 9.9 \times 10^{-5}$ $(\Delta k/k)/°F$ depending on the choice of fuel [2]. The MSRE's three control rods have a combined worth of 5.6 - 7.6 % $\Delta k/k$, again depending on the fuel [2].

	Reactivity, % Δk/k
Loss of delayed neutrons by circulating fuel	−0.3
Entrained gas	−0.2
Power coefficient (from rise in graphite temperature)	−0.1
Xenon poisoning (steady state at 10 Mw)	−0.7
Samarium-149 transient	−0.1
Burnup (120 g of fuel)	−0.1
Margin for operation of control rods	−0.4
Total	−1.9
Uncertainty in estimates (primarily xenon)	(+)−1.0 to (−)−0.4
Total	−2.9 to −1.5

Figure 4: Losses in reactivity associated with different phenomena [2].

3 MSRE adaption for modelling

3.1 Kophazi article review

Kophazi developed a 3D time-dependent simulation that was validated against MSRE data [6]. The model consists of a conventional neutronics model extended to deal with the precursor drift that is unique to MSRs and the core's thermal behavior. The thermal model includes the true geometry of the fuel channels as well as the moderator connecting them. Salt flow is assumed to be only in the reactor's axial direction, which reduces the number of non-linear variables that have to be solved for.

3.1.1 Neutronics

Neutrons are described with time-dependent multi-group diffusion theory as shown in eq. (1):

$$\frac{1}{v_g}\frac{\partial \phi_g}{\partial t} - \nabla \cdot D_g \nabla \phi_g + \Sigma_g^r \phi_g = \sum_{g \neq g'}^{G} \Sigma_{g' \to g}^s \phi_{g'} + \chi_g^p \sum_{g'=1}^{G} (1-\beta)\nu \Sigma_{g'}^f \phi_{g'} + \sum_{i}^{I} \lambda_i \chi_g^{d,i} C_i \tag{1}$$

Delayed neutron precursors are described by eq. (2):

$$\frac{\partial C_i}{\partial t} = \sum_{g'=1}^{G} \beta_i \nu \Sigma_{g'}^f \phi_{g'} - \lambda_i C_i - \frac{\partial}{\partial z} u C_i \tag{2}$$

with the last term representing the effect of fuel advection. In [6], eq. (1) is solved only in the reactor core, whereas eq. (2) is solved both in the core and in the MSRE external primary circuit. The external loop is modeled as a single pipe with a volume equal to the total volume of all the external components and pipelines of the primary circuitry [6]. It is not entirely clear how pipe cross sectional areas are handled, which would determine the transit times of individual pieces of fluid and subsequently the distribution of delayed neutron production in the external primary circuit. That is a piece that would likely need to be included in detailed Advanced Reactors and Fuel Cycles (ARFC) accident models.

As pointed out in [6], it is not possible to solve eqs. (1) and (2) on the same mesh. Equation (1) needs to be solved over the entire reactor cross-section including fuel channels and the graphite moderator in between, whereas eq. (2) is only applicable within the fuel channels. Kophazi solves this issue by introducing a fuel fraction parameter, p_f, and creating effective fuel flow channels. Figure 5 shows how Kophazi transforms the MSRE unit cell into neutronic and precursor elements. The number of effective fuel channels in the unit cell is determined by the refinement of the neutron grid, e.g. the # of channels is equivalent to the number of neutronic elements in a given MSRE unit cell. Consequently, once the domain is discretized (Kophazi uses the finite volume method), the delayed neutron source term (last term in eq. (1)) is multiplied by the fuel fraction p_f. Homogenization of the removal, scattering, and fission cross sections allows translation of the model from the real to effective geometry. Details of the discretization can be found in [6]. Unfortunately, neither [6] or cited references for the DALTON code, describe the boundary conditions used for neutrons.

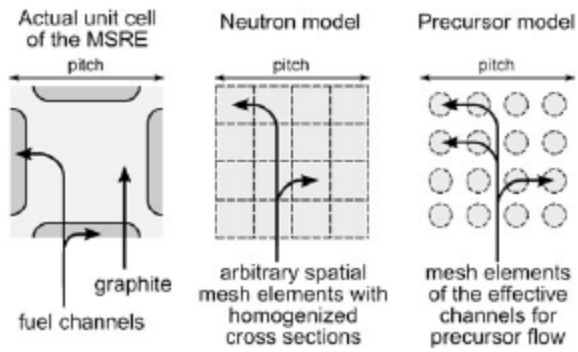

Figure 5: Homogenized model of fuel channels [6]

3.1.2 Thermal Hydraulics

Temperature in the fuel is determined by a 1-D convection model in the axial direction. Moderator temperatures are calculated with a 3-D conduction model. Coupling between the moderator and fuel temperatures is accomplished through an axially varying heat transfer coefficient calculated with a Nusselt correlation; see [6] for details of the Nusselt correlation. Thermal expansion of the fuel was not accounted for while determining the reactor temperature profile; however, expansion is considered ($\kappa = 0.57 mg/cm^3 K$) when determining homogenized neutron cross-sections. A model for the heat exchanger is included with a heat sink term proportional to the temperature difference between fuel and coolant salts, with the constant of proportionality determined from steady-state MSRE operational data. With the exception of the heat exchanger, the thermal code assumes zero heat flux boundary conditions.

3.1.3 Reactor Model

Cell weighting and cross section homogenization calculations were carried out using SCALE. In order to work with the 1D nature of the SCALE/CSAS/XSDRN code, the actual MSRE geometry was simplified as shown in fig. 6 where the actual geometry is on the left and the model geometry is on the right. Note that the far left sub-figure of fig. 5 is the zoomed, unit cell version of the left sub-figure of fig. 6.

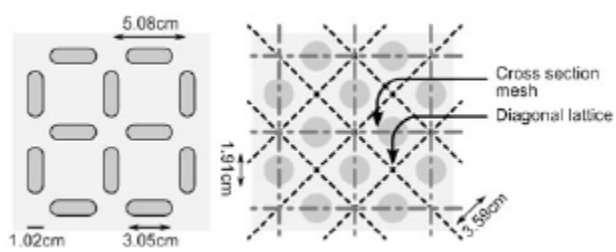

Figure 6: Left figure shows snapshot of actual MSRE core geometry; to satisfy the 1D SCALE/CSAS/XSDRN code, the more regular geometry in the right figure is used [6]

Neutron cross sections were collapsed into eight energy groups, distributed evenly on a logarithmic scale, with additional groups added in resonance and thermal regions. Control rods were modeled with albedo boundary conditions for thermal neutrons. More control rod modeling details can be found in [6].

Neutron and thermal calculations are performed on separate grids. The axial and horizontal neutron grids are both shown in fig. 7. It includes regions for the lower and upper salt plena, downcomer, and three control rods. A section of the horizontal thermal grid is shown in fig. 8. In general, the thermal grid is

finer than the neutron grid. Thermal and neutronic calculations are coupled in the following way. Each cell of the neutronics mesh (containing both fuel and moderator, recall fig. 5) is considered to have uniform cross-sections. To generate these constant cross-sections in each neutronic cell, the fine-mesh temperature fields from the thermal calculation (called THERM) are spatially averaged using an auxiliary program. This averaged neutron-cell-specific temperature is then used in conjunction with cross section interpolation tables generated by SCALE to determine the macroscopic cross sections for each neutron cell in the simulation.

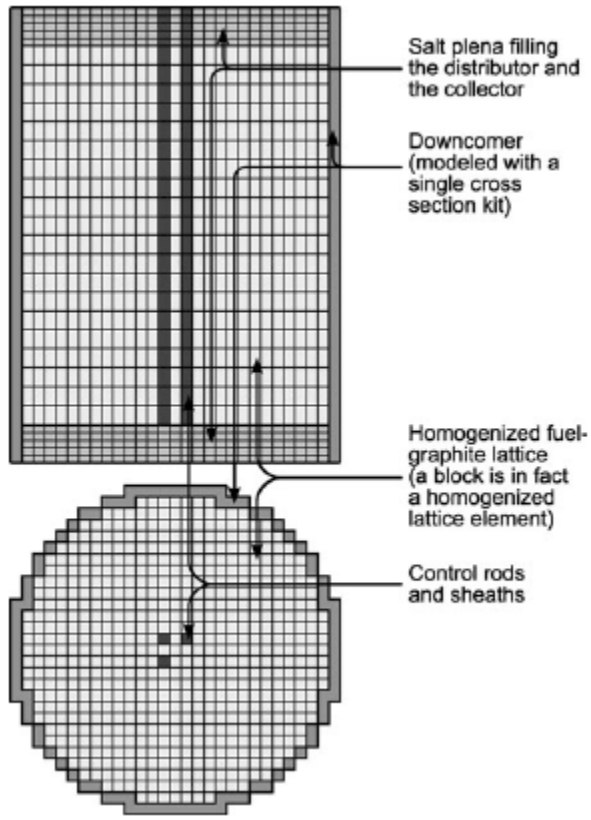

Figure 7: Grid for cross section homogenization and neutron cross sections [6]

Feedback from the neutron to thermal calculations is accomplished by translating the power distribution from the coarse neutron grid to the fine thermal grid using another auxiliary program. This power distribution is then used as a heat source in the thermal code.

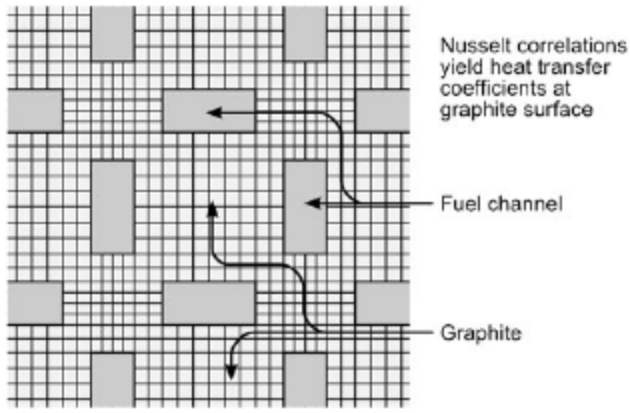

Figure 8: Cross-section of thermal grid [6]

3.2 ARFC MSRE Model Design Plan

1. Test whether `gmsh` can generate heterogeneous 3D mesh structure like that in the actual MSRE geometry.
2. For expected range of temperatures and some defined sequence of temperature step sizes, generate table of few-group neutronics cross sections with `SERPENT`.
   - If `gmsh`, or that failing `cubit`, is capable of generating heterogeneous grid, we can generate unique cross regions for each region
   - If we cannot generate heterogeneous grid, we will have to homogenize the cross sections similar to what is done in [6].
3. Armed with tables for quick interpolation of cross sections, we should be able to perform coupled calculations for neutron density and temperature.
   - As a first pass, the fuel salt flow velocity can be a user input as opposed to a non-linear variable that is a function of temperature.
   - However, the natural convection impacts on salt flow will significantly contribute to reactor dynamics and the ultimate goal will be to couple neutron flux and flow directly via temperature.
   - We will need to decide whether neutron and temperature calculations should be performed on the same grid, and in transient calculations whether those two fields should be updated on the same step-size.
     - If not using the same grid and/or step-size, the neutronic and temperature calculations will need to be split into their own applications and coupling will have to be performed using the MOOSE multi-app system.

3.3 ARFC MSRE Model Design Execution

3.3.1 Interpolation table generation

Based on the operating temperatures in table 1, initial interpolation tables will range from 824 to 1055 K for the fuel salt temperature, and from 858 to 1088 K for the moderator temperature. The minimum fuel salt temperature is equal to the heat exchanger inlet coolant salt temperature, while the maximum fuel salt temperature is 100 K greater than the maximum fuel salt core outlet temperature. The moderator limits are 60 °F greater than their fuel counterparts based on the experimental observation by Robertson that

moderator temperatures are only ever about 60 °F above the adjacent fuel. Reference temperatures for the fuel and moderator will be 922 K (the average of reactor inlet and outlet temperatures) and 953 K (graphite core average) respectively. These simulation set-points are summarized in table 2.

Serpent Simulation Temperatures	Value (K)
Fuel minimum	824
Fuel maximum	1055
Fuel reference	922
Moderator minimum	858
Moderator maximum	1088
Moderator reference	953

Table 2: Relevant temperature parameters for Serpent simulations

The fuel salt density dependence on temperature is based on the Fluoride-Lithium-Beryllium (FLiBe) density correlation in [7]:

$$\rho(kg/m^3) = 2413.03 - 0.4884 \cdot T(K) \tag{3}$$

This does not account for the change in density introduced by dissolution of fuel components in the salt. At the average fuel salt temperature in the reactor, the FLiBe correlation predicts a density of 1.96 grams/cm^3. This differs from the measured values in the MSRE of 2.24, 2.08, and 2.14 grams/cm^3 for thorium-uranium, highly-enriched uranium, and partially-enriched uranium fuel salts respectively. Moderator density is currently considered to be constant and equal to 1.843 grams/cm^3 [8].

Group constants required for eigenvalue calculation:

$$D_g, \Sigma_{rg}, \Sigma_{sg'g}, \chi_g, \nu_g \Sigma_{fg}, \kappa \tag{4}$$

Additionally, v_g is required for transient calculations.

4 MSBR

The most detailed report on ORNL's molten salt breeder reactor is the technical report by Robertson [9]. A nice overview of the MSBR geometry is shown in figs. 9 to 11.

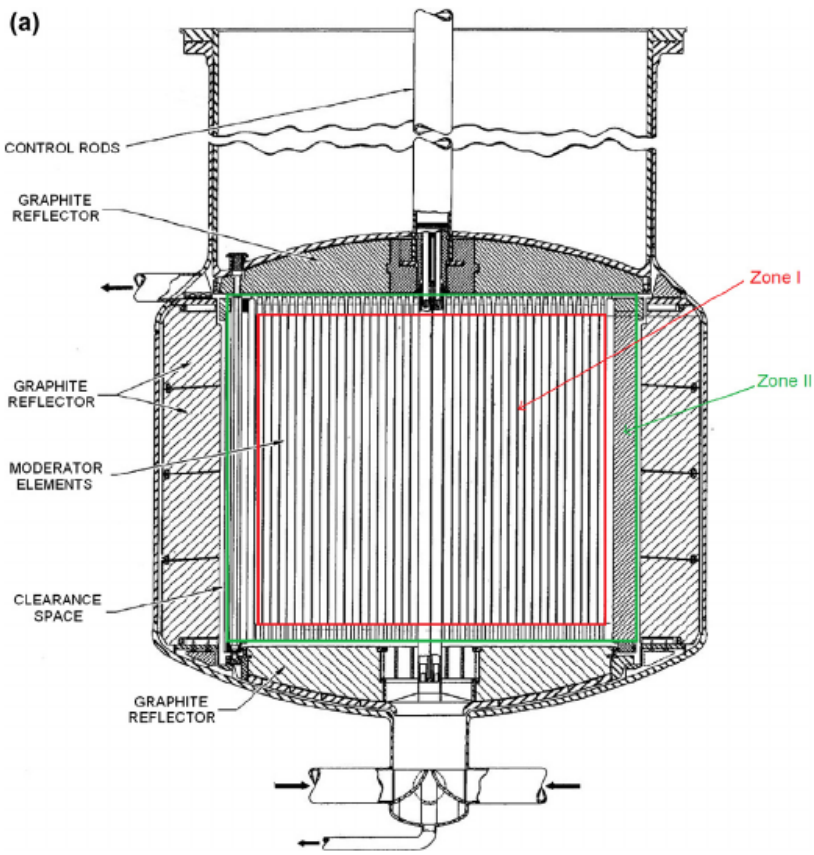

Figure 9: Vertical cross section of MSBR.

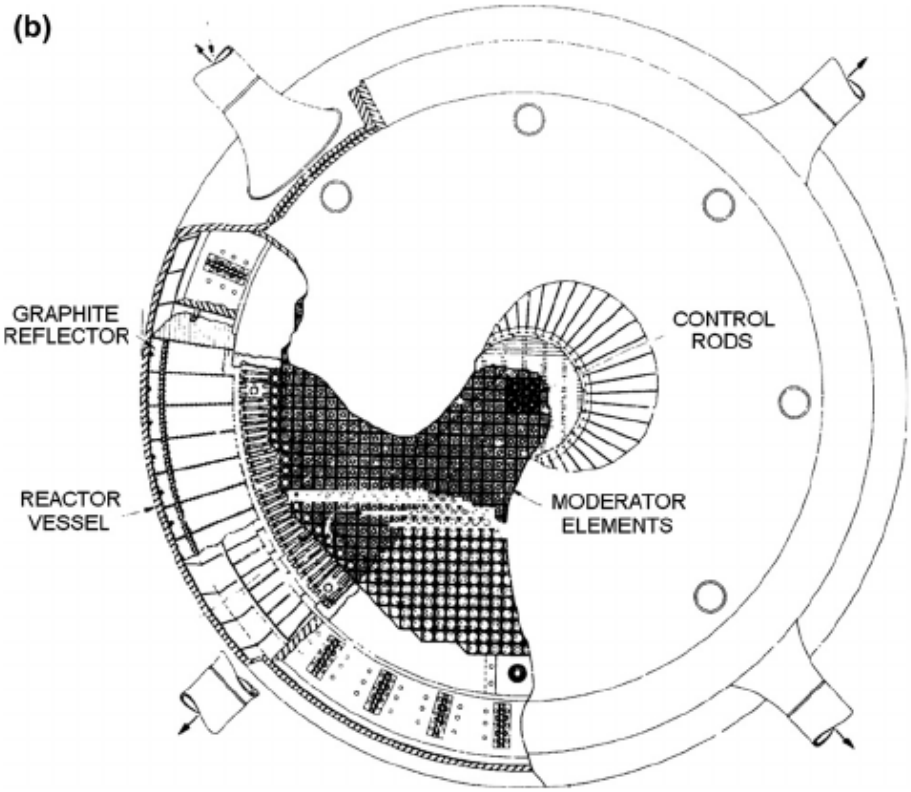

Fig. 2 (*continued*)

Figure 10: Horizontal cross section of MSBR.

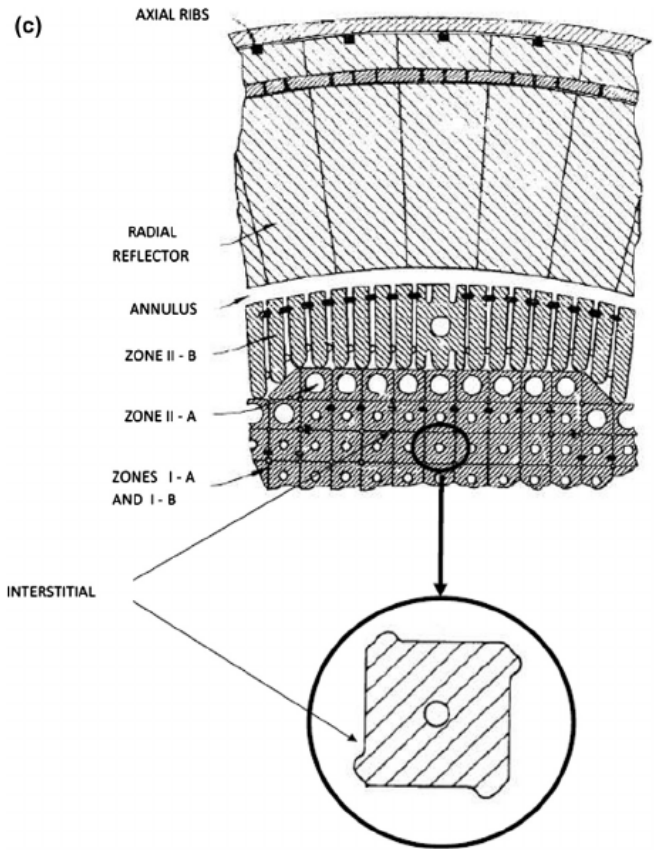

Figure 11: Zoomed in horizontal cross-section of MSBR. Shows presence of interstitials between graphite block elements in zone 1 of the reactor.

5 Simplified models

5.1 MSBR children

5.1.1 Cammi et. al.

This model is based on the multi-physics model by [8].

Model's simplifications:

1. Considered only one channel (diameter 0.05 m)
2. Hollow cylinder instead of block with a hole (with keeping the same fuel to graphite ratio)
3. Infinite length of the channel (z tends to infinity)

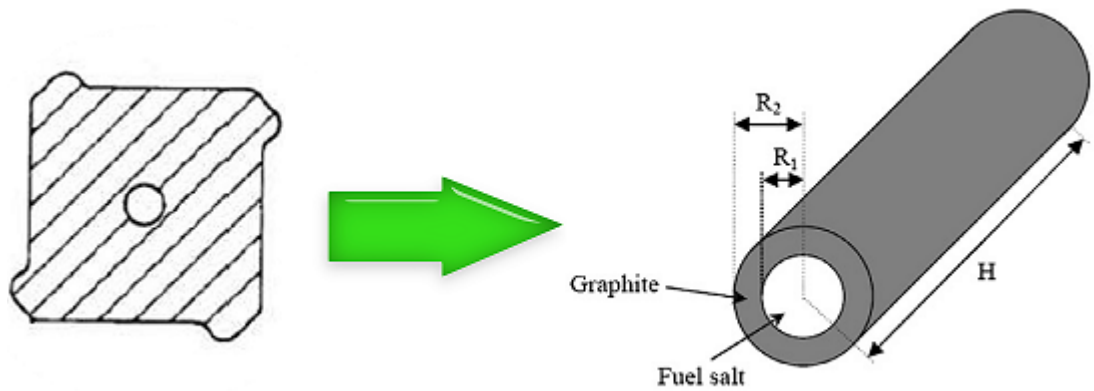

Figure 12: Cammi's simplification of MSBR's geometry [8].

Neutronics Boundary Conditions.

1. *Inlet boundary.* The "vacuum" condition is adopted for the fast and thermal neutron fluxes. The circulation of delayed neutron precursors is taken into account considering that a certain amount of DNPs can return into the core from the external loop:

$$C_{i,in}(t) = C_{i,out}(t - \tau_{EL}) \exp(-\lambda_i \tau_{EL})$$

2. *Outlet boundary.* "Vacuum" condition for fast and thermal neutrons.
3. *Wall and central boundaries.* Symmetry boundary conditions are imposed for the neutron fluxes at r=0 and $r = R_2$.

Flow and heat transfer boundary conditions.

1. *Inlet boundary.* The inlet velocity is given as $u_z = U_{in} = 1.47m/s$ and $u_r = 0$. Turbulent kinetic energy $k_{in} = 0.007m^2/s^2$. Turbulent dissipation rate $\epsilon_{in} = 0.033m^2/s^2$. $T_F = T_in = 839K$.
2. *Outlet boundary.* The "local one-way method" is exploited for u,k,ϵ, and T_F.

3. *Central boundaries.* Symmetry conditions are considered on the axis of the channel.

4. *Wall and central boundaries.* "Wall function approach" that use empirically-based relations in turbulent flows near solid boundaries in order to describe the velocity and temperature profile in the thin boundary layer near the wall, instead of solving the turbulence equations.

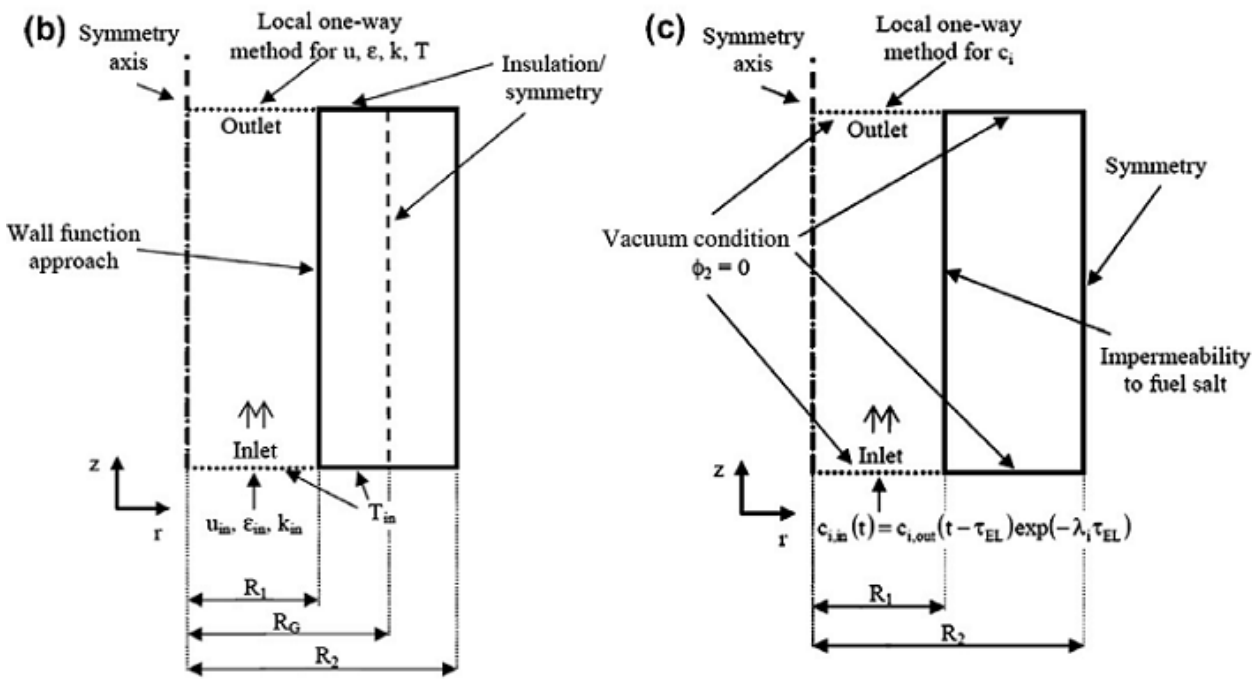

Figure 13: (b) Fluid flow and heat transfer boundary conditions; (c) neutron flux and DNP boundary conditions [8].

Acronyms

ARE Aircraft Reactor Experiment. 1

ARFC Advanced Reactors and Fuel Cycles. 7, 10

FLiBe Fluoride-Lithium-Beryllium. 11

gpm gallons per minute. 5

INL Idaho National Laboratory. 1

LMFBR Liquid-Metal-cooled Fast Breeder Reactor. 1

MSR Molten Salt Reactor. 1

MSRE Molten Salt Reactor Experiment. 1, 3–5, 7, 8, 10, 11

ORNL Oak Ridge National Laboratory. 1, 11

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